

Functional Annotation Report: PP_0581 (Q88QB3) — A High-Molecular-Weight FabG β -Ketoacyl-Thioester Reductase in *Pseudomonas putida* KT2440

Summary

PP_0581 (UniProt Q88QB3) encodes a soluble, cytoplasmic, NAD(H)-dependent β -ketoacyl (3-oxoacyl) thioester reductase of the short-chain dehydrogenase/reductase (SDR) superfamily. The catalyzed reaction is the stereospecific reduction of a 3-oxoacyl (β -ketoacyl) thioester to the corresponding (R)-3-hydroxyacyl thioester, using NADH as the hydride donor. Critically, this protein is **NOT** the essential housekeeping type-II fatty acid synthase (FAS-II) FabG that performs the ketoreduction step of every fatty-acid elongation cycle. That housekeeping role in *P. putida* KT2440 belongs to the separate, canonical 246-residue enzyme PP_1914, encoded in the classical *fabD*–*fabG*–*acpP* operon.

Instead, PP_0581 is a **high-molecular-weight FabG (HMwFabG, also called the FabG4-type)** — a 457-residue, ~47.8 kDa protein roughly double the size of a canonical FabG. It has a diagnostic two-domain architecture: an N-terminal "flavodoxin-type" domain fused to a C-terminal SDR ketoreductase (oxoreductase) domain. This architecture is the structural hallmark of the phylogenetically distinct HMwFabG group prototyped by *Mycobacterium tuberculosis* FabG4 (Rv0242c). An AlphaFold model (AF-Q88QB3-F1, global mean pLDDT 94.4) confirms a confidently folded two-domain protein with a geometrically competent SDR catalytic triad (Ser350–Tyr363–Lys367) and an NAD-selective aspartate (Asp251) cofactor-specificity determinant, consistent with the NADH preference characteristic of the FabG4/HMwFabG class.

Functionally, PP_0581 sits at the head of a **compact, self-contained, AraC-regulated acyl-CoA/ β -oxidation-type metabolic module**. Genomic-context analysis places it in a tightly spaced, co-directional operon (PP_0581–PP_0580–PP_0579) that is divergently transcribed from a thiolase gene (PP_0582) and controlled by an adjacent AraC-family regulator (PP_0583). Its immediate neighbor PP_0580 is a MaoC-domain (R)-specific enoyl-CoA hydratase — the enzyme

family that channels (R)-3-hydroxyacyl-CoA between β -oxidation and polyhydroxyalkanoate (PHA) metabolism in *Pseudomonas*. The absence of a co-localized transporter or acyl-CoA synthetase indicates the physiological substrate is an **intracellular CoA-thioester intermediate**, and the dedicated AraC regulator implies an inducible, substrate-specific catabolic function rather than constitutive housekeeping fatty-acid synthesis. **Confidence is high** for the enzyme chemistry, family, cofactor class, and cytoplasmic localization (structure/sequence/genomic evidence), and **moderate** for the exact physiological substrate, which has not been experimentally determined and remains the principal knowledge gap.

Target Identity Verification

Property	Value
UniProt	Q88QB3 (Q88QB3_PSEPK), TrEMBL, unreviewed
Locus	PP_0581 (OrderedLocusNames); GenBank AAN66208.1
Organism	<i>Pseudomonas putida</i> KT2440 (ATCC 47054 / DSM 6125 / CFBP 8728 / NCIMB 11950); TaxID 160488
Length / MW	457 aa / ~47.8 kDa
Genome position	chromosome, complement(677,644–679,017)
KEGG orthology	K00059, 3-oxoacyl-[ACP] reductase, EC 1.1.1.100
Family	SDR superfamily; NCBIfam PRK08261 (HMwFabG model)
Domains	Two NAD(P)-binding Rossmann domains (Gene3D 3.40.50.720 $\times 2$); C-terminal ketoreductase (SMART, res 221–394); Pfam adh_short_C2 (PF13561); PROSITE ADH_SHORT (PS00061)
PANTHER	PTHR42760:SF78 "3-oxoacyl-[ACP] reductase [NADH]"

The gene symbol, organism, protein family, and domain content in the provided UniProt record are internally consistent and were confirmed against KEGG, InterPro/Gene3D, and NCBIfam. "PP_0581" is an unambiguous ordered-locus name specific to KT2440, so there is no

cross-organism symbol ambiguity. The only nuance is that the generic annotation "3-oxoacyl-ACP reductase / FabG" **understates** the protein's true identity as a *high-molecular-weight* FabG paralog.

Key Findings

Finding 1: PP_0581 is a high-molecular-weight FabG (HMwFabG/FabG4-type), not the housekeeping FAS-II FabG

PP_0581 is 457 amino acids long (~47.8 kDa), roughly double the length of a canonical FabG (~245 residues, ~25 kDa). This size difference reflects a fundamentally different domain organization. Structural-domain classifiers (Gene3D) assign **two** NAD(P)-binding Rossmann-fold domains (CATH 3.40.50.720 × 2): an N-terminal domain spanning roughly residues 1–215 and a C-terminal SDR ketoreductase domain (SMART ketoreductase 221–394; Pfam adh_short_C2, PF13561). This tandem "N-terminal flavodoxin-type domain + C-terminal oxoreductase domain" arrangement is the defining, diagnostic feature of the HMwFabG group.

The HMwFabGs form a phylogenetically separate clade distinct from the low-molecular-weight housekeeping FabGs. As established for the *Mycobacterium tuberculosis* prototype FabG4, "HMwFabGs (high-molecular-weight FabG) form a phylogenetically separate group of FabG enzymes. FabG4, an HMwFabG from *Mycobacterium tuberculosis*, contains two distinct domains, an N-terminal 'flavodoxin-type' domain and a C-terminal oxoreductase domain" (PMID: 23163771). PP_0581 reproduces exactly this architecture. Independent classifiers corroborate the assignment: PANTHER places PP_0581 in PTHR42760:SF78, "3-oxoacyl-[ACP] reductase [NADH]," and the NCBIfam model PRK08261 — the specific HMwFabG model — is matched.

Importantly, HMwFabGs are structurally and cofactor-distinct from the canonical enzymes. Work on *Acinetobacter baumannii* reductases describes the "high-molecular weight 3-oxoacyl-ACP reductase (HMwFabG), showing differences in structure and coenzyme preference" (PMID: 33782435), supporting the conclusion that PP_0581 does not fulfill the housekeeping FabG role.

The genuine housekeeping FAS-II FabG in KT2440 is a separate protein: the 246-residue PP_1914, encoded in the classical fatty-acid synthesis operon (PP_1913 *fabD*, malonyl-CoA:ACP transacylase / PP_1914 *fabG* / PP_1915 *acpP*, acyl carrier protein). KT2440 carries five K00059-

annotated (3-oxoacyl-ACP reductase) paralogs — PP_0581, PP_1852 (249 aa), PP_1914 (246 aa), PP_2540 (258 aa), and PP_2783 (245 aa) — but only PP_0581 is high-molecular-weight and genomically remote from the *fab* operon, further distinguishing it from the pack.

Finding 2: PP_0581 has intact SDR catalytic machinery — a predicted NAD(H)-dependent β -ketoacyl reductase, cytoplasmic

The C-terminal SDR domain of PP_0581 contains all the canonical sequence motifs required for a functional short-chain dehydrogenase/reductase. The cofactor-binding Rossmann glycine motif (TGxxxGxG) is present as **V-T-G-A-A-R-G-I-G at residues 225–233**, and the SDR catalytic Tyr-x-x-x-Lys motif appears as **Y363-A-A-S-K367**. The upstream catalytic serine is in the ~S350/S352 region, completing the canonical SDR **Ser–Tyr–Lys catalytic triad**. The PROSITE ADH_SHORT signature (PS00061) and the Sc_DH/Rdtase conserved-site pattern (IPR020904) are both matched.

By homology to the FabG4 prototype — whose catalytic triad is Ser347/Tyr360/Lys364 — PP_0581 is predicted to reduce a β -ketoacyl (3-oxoacyl) thioester to the corresponding (R)-3-hydroxyacyl thioester using NAD(P)H, belonging to the EC 1.1.1.100 reaction family (oxidoreductase acting on the CH–OH/keto group, GO:0016616). The direct enzymatic characterization of FabG4 establishes both the reaction and the mechanism: "The enzyme exhibits ketoacyl CoA reductase activity by reducing acetoacyl CoA to 3-hydroxyacyl CoA in presence of NADH. Conserved catalytic triad Ser347, Tyr360, and Lys364 constitute the active site residues of the KAR domain" ([PMID: 21081168](#)).

Two additional features refine the substrate and cofactor predictions. First, the FabG4 prototype accepts both CoA- and ACP-based substrates: "Structural data ACP inhibition studies indicate that FabG4 can accept both CoA- and ACP-based fatty-acyl substrates" ([PMID: 23163771](#)). Second, the NADH cofactor preference is a defining feature of the class: "The enzyme requires NADH for β -ketoacyl reductase activity" ([PMID: 22750865](#)), consistent with PP_0581's PANTHER "[NADH]" subfamily assignment.

Regarding localization, PP_0581 carries no annotated signal peptide and no predicted transmembrane region, consistent with a **soluble cytoplasmic enzyme** — the compartment where both fatty-acid and acyl-CoA metabolism occur.

Findings 3 & 5: PP_0581 heads a compact, AraC-regulated acyl-CoA/ β -oxidation-type module

Genomic-context analysis provides the strongest available evidence for the physiological role of PP_0581. The gene sits within a functionally coherent cluster on the KT2440 chromosome (around 676–681 kb):

Locus	Product / annotation	Function in module
PP_0579	conserved hypothetical protein	co-operonic; unknown accessory
PP_0580	MaoC-domain protein	(R)-specific enoyl-CoA hydratase
PP_0581	HMwFabG reductase (this protein)	β-ketoacyl-CoA $\rightarrow$ (R)-3-hydroxyacyl-CoA
PP_0582	thiolase (acetyl-CoA C-acetyltransferase, K00626, EC 2.3.1.9)	thiolytic C–C cleavage / condensation
PP_0583	AraC-family transcriptional regulator	inducible, substrate-specific control

Precise coordinate analysis shows that PP_0581 (–strand, 677,644–679,017), PP_0580 (–strand, 676,790–677,641; only a 2-bp gap to PP_0581), and PP_0579 (–strand, 676,407–676,805; a 16-bp overlap with PP_0580) form a co-directional, tightly spaced operon transcribed in the order PP_0581 $\rightarrow$ PP_0580 $\rightarrow$ PP_0579. The thiolase PP_0582 (+strand, 679,114–680,391) is **divergently transcribed** just 96 bp from the PP_0581 start, and the AraC-family regulator PP_0583 (+strand, 680,481–681,260) lies immediately beyond it — the classic layout of an AraC-controlled divergent catabolic locus.

The wider genomic neighborhood is functionally unrelated — PP_0578 is a methyltransferase, PP_0584 is a chemotaxis methyl-accepting protein (MCP), and PP_0585–PP_0588 handle metal/copper homeostasis — so the metabolic module is confined to PP_0579–PP_0583.

The enzymatic logic of this cluster is decisive. A **thiolase + (R)-specific hydratase + β -ketoacyl reductase** trio is the enzymatic hallmark of a CoA-dependent β -oxidation-type / (R)-3-hydroxyacyl-CoA metabolic module. In *Pseudomonas*, MaoC-family (R)-hydratases specifically bridge β -oxidation and PHA biosynthesis: "MaoC is a new enoyl-CoA hydratase involved in supplying (R)-3-hydroxyacyl-CoA from the beta-oxidation pathway to PHA biosynthetic pathway" (PMID: 12949091). The presence of PP_0580 as a co-operonic (R)-hydratase strongly defines the CoA-thioester pathway context of the PP_0581 reductase.

Two negative observations sharpen the interpretation. First, no cognate transporter or acyl-CoA synthetase is co-localized with the module, indicating the substrate is an **intracellular CoA-thioester intermediate** rather than an imported or newly activated fatty acid. Second, the dedicated adjacent AraC regulator implies an **inducible, substrate-specific catabolic operon** rather than constitutive housekeeping fatty-acid synthesis — reinforcing that PP_0581 is not the housekeeping FabG.

Finding 4: AlphaFold confirms a folded two-domain HMwFabG with a competent SDR catalytic triad

The AlphaFold DB model AF-Q88QB3-F1 (v6) covers all 457 residues at very high confidence: **global mean pLDDT = 94.4, with 98% of residues at pLDDT > 70**. Crucially, both structural modules are well-folded rather than one being a disordered extension: the N-terminal domain (residues 1–215) has mean pLDDT 91.9, and the C-terminal SDR ketoreductase domain (216–457) has mean pLDDT 96.6. This directly confirms a genuine two-domain (N-terminal flavodoxin-type + C-terminal oxoreductase) HMwFabG architecture rather than a disordered N-terminal extension.

In the modeled active site, the catalytic triad is geometrically competent and adopts canonical SDR proton-relay geometry: **Tyr363-OH...Lys367-NZ = 4.29 Å** and **Ser350-OG...Tyr363-OH = 4.24 Å**. The YxxxK motif (Y363-AAS-K367) and the cofactor glycine-rich motif (VTGAARGIG, residues 225–233) are both positioned within the same C-terminal domain, and the conserved SDR asparagine (H-N-A-G, ~residues 301–304) is also present — consistent with a fully assembled, catalytically poised active site. This upgrades the catalytic assignment from motif-matching to structure-supported.

Finding 6: An NAD-selective aspartate (Asp251) supports NADH over NADPH preference

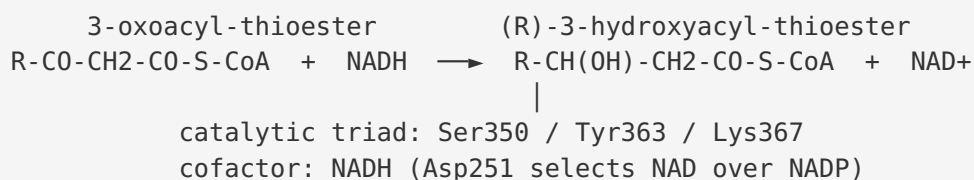
The cofactor specificity of SDR/Rossmann enzymes is dictated by the residue at the C-terminus of the β B strand, roughly 18 residues downstream of the glycine-rich motif. In PP_0581 this position is an acidic aspartate — **Asp251** (in the context ...DVADVLLLD**251**VPQ...). An acidic residue at this canonical position hydrogen-bonds the 2'- and 3'-hydroxyls of the adenosine ribose and is the classical fingerprint of **NAD(+)-specific** SDR enzymes. NADP-specific enzymes instead carry a basic residue and a 2'-phosphate-binding pocket. The presence of Asp251 therefore predicts NAD(H) rather than NADPH dependence.

This structural determinant is fully concordant with two independent lines of evidence: the PANTHER subfamily classifier PTHR42760:SF78 ("3-oxoacyl-[ACP] reductase [NADH]") and the experimentally established NADH dependence of the FabG4/HMwFabG prototype ([PMID: 22750865](#)). It also further distinguishes PP_0581 from the typically NADPH-using housekeeping FAS-II FabG. The other conserved SDR residues — the N-A-G asparagine motif (~residues 301–304), catalytic Ser350, Tyr363, and Lys367 — are all present, completing a canonical, NADH-selective SDR active site.

Mechanistic Model / Interpretation

The reaction

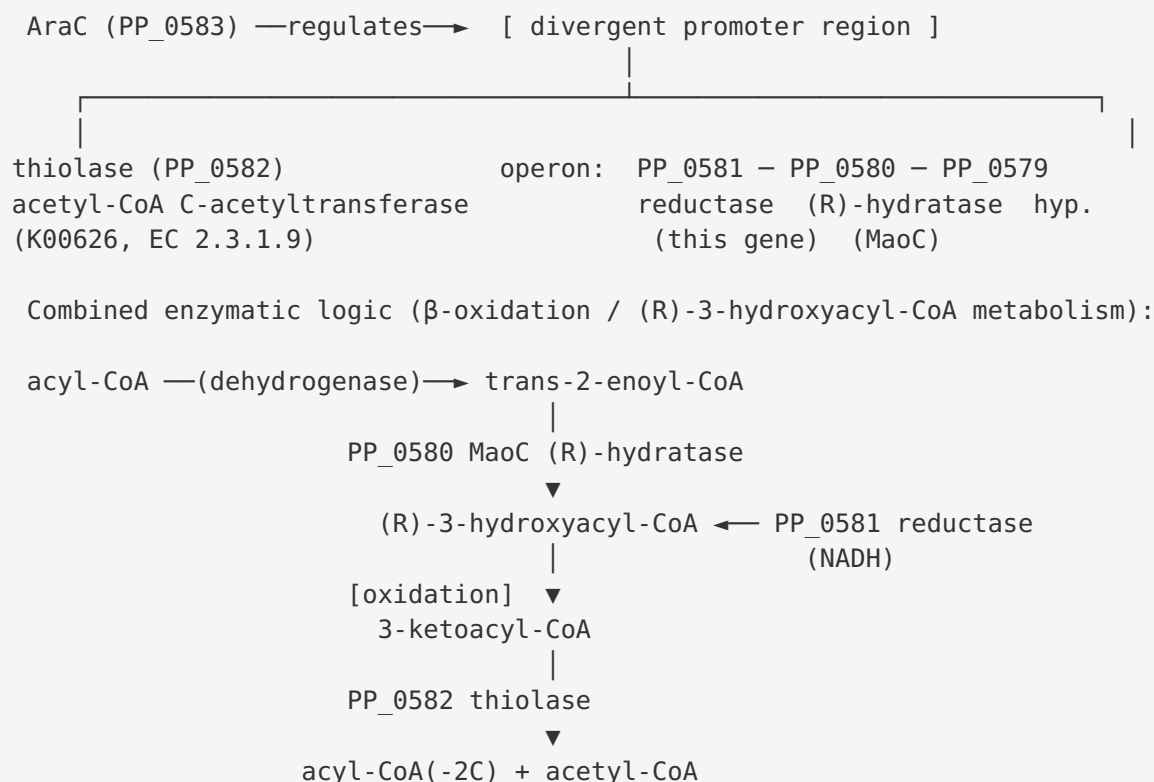
PP_0581 catalyzes the NADH-dependent ketoreduction step of a CoA-thioester (and possibly ACP-thioester) metabolic cycle:



Mechanistically, in the SDR proton relay: catalytic Tyr363 donates a proton to the developing alkoxide of the reduced substrate; Lys367 lowers the pKa of Tyr363 and binds the nicotinamide ribose; and Ser350 stabilizes and orients the substrate carbonyl. The AlphaFold model places these residues in canonical distance geometry (Tyr–Lys 4.29 Å, Ser–Tyr 4.24 Å), indicating a poised, functional active site.

The pathway context

The genomic module assembles a complete CoA-thioester processing unit:



The thiolase + (R)-hydratase + NADH-reductase combination is diagnostic of a CoA-dependent β -oxidation-type module. In *Pseudomonas*, the MaoC (R)-hydratase specifically supplies (R)-3-hydroxyacyl-CoA — a direct precursor for medium-chain-length polyhydroxyalkanoate (PHA) synthesis — linking this module to the intersection of fatty-acid β -oxidation and PHA carbon-storage metabolism. PP_0581's NADH-dependent ketoreductase activity provides the reductive step that generates or interconverts the 3-hydroxyacyl-CoA/3-ketoacyl-CoA pool.

Two features mark this as a **dedicated, inducible catabolic module** rather than housekeeping biosynthesis: (1) the standalone AraC regulator implies substrate-inducible expression, and (2) the lack of a co-localized transporter or acyl-CoA synthetase indicates the substrate is generated intracellularly as a CoA-thioester intermediate. This is entirely consistent with PP_0581 being an HMwFabG — a class functionally and phylogenetically separate from the housekeeping FAS-II FabG (PP_1914), which handles constitutive membrane-lipid fatty-acid synthesis.

Why PP_0581 is not the housekeeping FabG — summary comparison

Property	PP_0581 (Q88QB3)	Housekeeping FabG (PP_1914)
Length	457 aa (~47.8 kDa)	246 aa (~25 kDa)
Domain architecture	Two domains (N-flavodoxin-type + C-oxoreductase)	Single SDR domain
Class	HMwFabG / FabG4-type	Low-molecular-weight canonical FabG
Genomic context	AraC-regulated acyl-CoA module (PP_0579–0583)	<i>fabD–fabG–acpP</i> FAS-II operon
Cofactor	NADH (Asp251 determinant)	typically NADPH
PANTHER	PTHR42760:SF78 "[NADH]"	canonical 3-oxoacyl-ACP reductase
NCBIfam model	PRK08261 (HMwFabG)	canonical FabG model
Role	Inducible CoA-thioester ketoreduction	Essential FAS-II elongation

Evidence Base

The functional model rests on a combination of homology to a well-characterized structural prototype (FabG4), bioinformatic domain/motif analysis, AlphaFold structural modeling, and genomic-context inference. No experimental study on PP_0581 itself was located; the assignment is therefore an evidence-based inference. Key supporting literature:

PMID	Title (abbrev.)	How it supports the findings
23163771	<i>Crystal structure of hexanoyl-CoA bound to β-ketoacyl reductase FabG4 of M. tuberculosis</i>	Defines the HMwFabG group and its diagnostic two-domain architecture (N-flavodoxin + C-oxoreductase) that PP_0581 shares; shows FabG4 accepts both CoA- and ACP-based substrates.
21081168	<i>Crystal structure of FabG4 from M. tuberculosis reveals importance of C-terminal residues</i>	Establishes the reaction (acetoacetyl-CoA $\rightarrow$ 3-hydroxyacyl-CoA, NADH) and the Ser347/Tyr360/Lys364 triad that PP_0581's C-terminal domain reproduces.
22750865	<i>Crystallization of high-molecular-weight ketoacyl reductase FabG4 complexed with NADH</i>	Confirms the NADH cofactor requirement characteristic of the HMwFabG class, matching PP_0581's Asp251 determinant and PANTHER "[NADH]" assignment.
33782435	<i>Insights into A. baumannii fatty acid synthesis 3-oxoacyl-ACP reductases</i>	Confirms HMwFabGs are structurally and cofactor-distinct from canonical FabG, supporting that PP_0581 is not the housekeeping enzyme.
12949091	<i>New enoyl-CoA hydratase in mcl-PHA biosynthesis</i>	Identifies the co-operonic PP_0580 MaoC-domain protein as an (R)-specific enoyl-CoA hydratase supplying (R)-3-hydroxyacyl-CoA from β -oxidation to PHA, defining the CoA-based pathway context.

Additional context papers illustrate the broader theme that FabG-like SDRs are functionally diverse and that database annotation alone can be misleading:

- **Annotation can be misleading:** [PMID: 31443964](#) describes an *A. baumannii* SDR annotated as "putative FabG" whose crystal structure showed a FabG-like fold but which was **not** active against the canonical FabG substrate acetoacetyl-CoA — a caution that structural hallmarks do not guarantee canonical substrate specificity. This directly motivates experimental verification of PP_0581's substrate.
- **Specialized FabG paralogs exist:** [PMID: 31560825](#) (FabG3 in *Xanthomonas* xanthomonadin biosynthesis) and [PMID: 33846444](#) (ApeQ in aryl-polyene biosynthesis) show that FabG-family reductases can be dedicated to specialized secondary-metabolite pathways rather than housekeeping FAS-II — consistent with PP_0581 being a specialized, AraC-regulated paralog.

- **The catalytic-triad concept has nuance:** [PMID: 33545175](#) reported that *E. coli* FabG mutants lacking the assigned Ser/Tyr/Lys triad residues retained activity, suggesting canonical FabG may not be a classical SDR. This tempers over-reliance on the triad alone but does not alter the overall reductase assignment for PP_0581.
- **Multiple functional OARs can coexist:** [PMID: 26975437](#) (*S. meliloti* NodG) demonstrates a bacterium can possess several 3-oxoacyl-ACP reductases capable of fatty-acid synthesis — consistent with KT2440's five K00059 paralogs.
- **Inhibitor validation of FabG4:** [PMID: 24129589](#) confirms FabG4 is a validated β -ketoacyl-CoA reductase whose inhibitors bind the NADH cofactor domain, reinforcing the KAR-domain functional assignment for the HMwFabG class.

Limitations and Knowledge Gaps

1. **No direct experimental study of PP_0581 exists.** All conclusions are inferred from homology (to *M. tuberculosis* FabG4), sequence-motif analysis, AlphaFold structural modeling, and genomic context. The gene has not been cloned, expressed, purified, or assayed in the reviewed literature.
2. **The precise physiological substrate is unknown.** While the module context indicates an intracellular CoA-thioester β -ketoacyl substrate, the chain length, degree of substitution, and whether ACP-linked substrates are also accepted have not been determined. The cautionary example of the *A. baumannii* SDR that failed to reduce acetoacetyl-CoA despite a FabG-like fold ([PMID: 31443964](#)) underscores that substrate specificity cannot be assumed from fold alone.
3. **Cofactor preference is predicted, not measured.** The Asp251 determinant and PANTHER classification strongly predict NADH, but this has not been confirmed enzymatically for PP_0581 specifically.
4. **The metabolic destination is inferred.** The link to β -oxidation and/or PHA metabolism is based on the MaoC (R)-hydratase neighbor and thiolase, but flux through this module and the inducer sensed by AraC PP_0583 are unknown.
5. **The function of PP_0579** (co-operonic conserved hypothetical protein) is undefined and could modify the module's activity or substrate range.

6. **The role of the N-terminal "flavodoxin-type" domain** in PP_0581 is not established. In FabG4 this domain is non-catalytic and contributes to oligomerization/substrate handling; its specific contribution in PP_0581 is unmodeled beyond the confident fold.

Proposed Follow-up Experiments / Actions

1. **Recombinant expression and enzyme assay.** Clone PP_0581, express in *E. coli*, purify, and assay β -ketoacyl reductase activity spectrophotometrically (NAD(P)H oxidation at 340 nm) against a panel: acetoacetyl-CoA, longer β -ketoacyl-CoAs (C6–C14), and β -ketoacyl-ACP. This tests the reaction, substrate specificity, and CoA-vs-ACP preference.
 2. **Cofactor-preference determination.** Measure kinetics with NADH vs NADPH to confirm the Asp251-predicted NAD(H) selectivity, and test directly via a D251→basic mutant that should switch specificity toward NADPH.
 3. **Catalytic-triad mutagenesis.** Generate S350A, Y363F, and K367A mutants and assay activity to confirm the SDR mechanism — while heeding the *E. coli* FabG precedent ([PMID: 33545175](#)) that residual activity is possible.
 4. **Operon/transcription mapping and AraC induction.** Use RT-PCR/RNA-seq to confirm the PP_0581–PP_0580–PP_0579 operon boundaries and the divergent PP_0582/PP_0583 arrangement, and identify the small-molecule inducer that activates the AraC regulator PP_0583.
 5. **Phenotypic gene-deletion analysis.** Construct a PP_0581 knockout in KT2440 and screen for growth phenotypes on defined carbon sources (fatty acids of varying chain length, aromatics, alkanoates), altered fatty-acid/PHA profiles, and stress conditions — to place the module in β -oxidation vs PHA metabolism.
 6. **In vitro reconstitution of the module.** Co-purify PP_0581, PP_0580 (MaoC hydratase), and PP_0582 (thiolase) and reconstitute the pathway with a defined acyl-CoA to establish the metabolic direction (degradative β -oxidation vs (R)-3-hydroxyacyl-CoA supply for PHA).
 7. **Structural validation.** Solve the crystal structure of PP_0581 $\pm$ NADH $\pm$ substrate to confirm the AlphaFold-predicted two-domain architecture and active-site geometry, and to superpose on FabG4 to quantify fold homology.
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Conclusion

PP_0581 (Q88QB3) is best annotated as a **soluble, cytoplasmic, NADH-dependent, high-molecular-weight FabG (HMwFabG/FabG4-type) β -ketoacyl (3-oxoacyl) thioester reductase** that catalyzes the reduction of a 3-oxoacyl-thioester to an (R)-3-hydroxyacyl-thioester. It is functionally and phylogenetically distinct from the essential housekeeping FAS-II FabG (PP_1914) and instead operates as the ketoreduction step of a compact, AraC-regulated (PP_0583) acyl-CoA/ β -oxidation-type module — a co-directional PP_0581–PP_0580 (MaoC (R)-hydratase)–PP_0579 operon divergent from a thiolase (PP_0582) — acting on an intracellular CoA-thioester intermediate at the intersection of fatty-acid β -oxidation and (R)-3-hydroxyacyl-CoA/PHA metabolism. This assignment is well-supported by structural homology, motif analysis, AlphaFold modeling, and genomic context, but awaits direct experimental confirmation of its substrate, cofactor, and physiological role.

References (PubMed)

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